MDMS Master Database Index

This database handles the core operational data for smart meter installations, consumer details, utility hierarchies, and the master lookup tables that define system statuses and categories.

Navigation Guide for Agents

1. Start here or in the root bundle index.
2. Always consult `relationships.md` before writing multi-table SQL.
3. Only join tables that appear in the documented joins.

Core Operational Tables

These are the primary transaction and entity tables. Note that Network and Organisation tables contain self-referencing hierarchy keys.

- **Consumer Lookup**
  - *Description:* Core consumer profiles, connectivity statuses, tariff categories, and installation timestamps.
  - *Keywords:* consumer, rrnumber, account.
- **Meter Lookup**
  - *Description:* Smart meter hardware details, physical properties, manufacturers, and logical mappings to networks/consumers/ organisation.
  - *Keywords:* meter, smart meter.
- **Organisation Lookup**
  - *Description:* Utility office hierarchy (HQ, Circle, Division, etc.) with self-referencing parent-child structure.
  - *Keywords:* organisation, hierarchy, office, division, section.
- **Network Lookup**
  - *Description:* Physical electrical grid hierarchy (Substation, Feeder, DTR) with self-referencing parent-child structure.
  - *Keywords:* network, substation, feeder, dtr, hierarchy.

Master Tables (Lookups)

These tables define the enums, categories, and configurations used by the core operational tables:

- Connection Status
    - *Description:* Defines physical connectivity (Connected, Disconnected, Permanent Disconnection).
- Connection Category
    - *Description:* Tariff and connection categorization (Domestic, Commercial, Agriculture, etc.).
- Payment Type Contract
    - *Description:* Payment modes (Prepaid vs. Postpaid).
- Sanctioned Load Type
    - *Description:* Units of measurement for electrical power (KW, KVA, HP, W).
- Device Manufacturer
    - *Description:* Complete list of authorized smart meter brands and manufacturers (L&T, Genus, Secure, etc.).
- HES
    - *Description:* Head-End Systems responsible for smart meter data collection.
- Meter Phase
    - *Description:* Physical phase configurations of meters (1 PH, 3 PH, HT).
- Network Hierarchy
    - *Description:* Tiers of the physical grid (Substation, Feeder, DTR).
- Organisation Hierarchy
    - *Description:* Structural levels of the utility (HQ, Circle, Division, Subdivision, Section).

Relationship Map

- Relationships – Canonical join graph and alias recommendations for mdms_master.

Table: l_consumer_lookup

Description

This table contains the primary information for consumers in the MDMS, including names, addresses, connectivity states, and physical attributes.

Columns

- **Consumer_TblRefID** (int): Unique internal identity number for each consumer.
- **MeterLookup_TblRefId** (int): FOREIGN KEY to `l_meter_lookup`. Maps the smart meter attached to the consumer.
- **RRNumber** (varchar): PRIMARY KEY. Unique identifier number used to identify the consumer.
- **Consumer_Name** (nvarchar): Name of the consumer.
- **Consumer_FatherName** (nvarchar): Name of the consumer's father.
- **ConnectionStatus_TblRefID** (smallint, Enum: `1`='connected', `2`='disconnected', `3`='permanent disconnected'): FOREIGN KEY to `m_connection_status`. Stores the connectivity state of the smart meter.
- **ConnectionCategory_TblRefID** (int): FOREIGN KEY to `m_connection_category`. Stores information about the consumer's tariff category.
- **Consumer_Address** (nvarchar): Consumer's physical address.
- **Consumer_Pincode** (varchar): Consumer's postal pincode.
- **Consumer_MobileNumber** (varchar): Consumer's mobile phone number.
- **Consumer_Email** (varchar): Consumer's email address.
- **Account_ID** (varchar): Unique installation number for the consumer.
- **Sanctioned_Load_KW** (decimal): Sanctioned electrical load of the consumer (in kW).
- **PaymentContract_TblRefID** (int, Enum: `1`='postpaid', `2`='prepaid'): FOREIGN KEY to `m_paymenttype_contract`. Defines the consumer's payment type.
- **Nearest_AccountID** (varchar): Nearest account ID. If prefixed with 'NSC', it denotes a New Service Connection.
- **SL_TYPE_ID** (smallint): FOREIGN KEY to `m_sl_type`. Unit of measurement for sanctioned load and contract demand.
- **Contract_Demand** (decimal): Contract demand of the consumer.
- **isPrevilege** (tinyint, Enum: `0`='No', `1`='Yes'): Indicates if the consumer has privilege status (exempt from disconnection).
- **Privilege_StartDate** (date): Start date of the privilege period (applicable when isPrevilege = 1).

- **Privilege_Enddate** (date): End date of the privilege period (applicable when isPrevilege = 1).
- **isStagger** (tinyint, Enum: `0` ='No', `1` ='Yes'): Indicates if the consumer has a staggered disconnection schedule.
- **Stagger_Start_date** (datetime): Start date of the stagger period (applicable when isStagger = 1).
- **Stagger_End_date** (datetime): End date of the stagger period (applicable when isStagger = 1).
- **Installment** (float): Installment amount to be deducted daily.
- **InstallmentStartDate** (date): Start date from which the daily installment deduction begins.
- **InstallmentEndDate** (date): End date when the installment deduction stops.
- **Installment_Received_DateTime** (datetime): Date and time when the latest installment details were received.
- **isForceDisconnect** (tinyint, Enum: `0` ='No', `1` ='Yes'): Indicates if the consumer is marked for forced disconnection.
- **ForceDisconnectDateTime** (datetime): Date and time when the consumer was force-disconnected (applicable when isForceDisconnect = 1).
- **EntryTimeStamp** (datetime): Timestamp when the consumer was created in the system.
- **isTOD** (tinyint, Enum: `0` ='No', `1` ='Yes'): Indicates if Time of Day (TOD) billing is enabled for the consumer.
- **MRU** (varchar): Name of the Meter Reading Unit to which the consumer belongs.
- **MasterSyncDate** (datetime): Date and time when the consumer master sync was received from the utility/discom.
- **RC_DC_DateTime** (datetime): Date and time when the consumer's smart meter was last connected (RC) or disconnected (DC).
- **ServiceDate** (datetime): Date and time when the consumer's first smart meter was installed.
- **isLRCF** (bit, Enum: `0` ='No', `1` ='Yes'): Indicates if Local Relay Connect Functionality is enabled (null is treated as No).
- **LRCF_UpdateDateTime** (datetime): Date and time when LRCF was enabled (applicable when isLRCF = 1).
- **Area_Type** (varchar, Enum: `NON-RAPDRP` , `RAPDRP`): Defines the area type of the consumer.

- **Connection_Type** (varchar, Enum: `Permanent`, `Temporary`): Defines the nature of the consumer's connection.
- **LoadType** (varchar, Enum: `SanctionLoad`, `ContractDemand`): Defines the load type category of the consumer.
- **Discom_Push_Date** (datetime): Date and time when the meter installation details were pushed to the utility/discom.
- **accntstatus_tblrefid** (int, Enum: `1`='regular (CD)', `2`='temporary disconnected (TD)', `3`='permanent disconnected (PD)'): Stores the current account status of the consumer.

Table: l_meter_lookup

Description

This table contains the hardware records for all smart meters in the system. It tracks the physical properties (manufacturer, phase, meter type), configuration (multiplying factor, net metering), and logical mappings to consumers, networks, and organizations.

Columns

- **MeterLookup_TblRefID** (int): PRIMARY KEY. Unique internal identity number for each smart meter.
- **Consumer_TblRefID** (int): FOREIGN KEY to `l_consumer_lookup`. Maps the smart meter to its attached consumer.
- **Meter_Serial_Number** (varchar): Unique serial number of the smart meter.
- **HESID** (smallint): FOREIGN KEY to `m_hes`. Maps the meter to its corresponding Head-End System (HES).
- **ServicePointMeterPhase_TblRefID** (int): FOREIGN KEY to `m_servicepoint_meterphase`. Stores the phase configuration of the smart meter.
- **DeviceManufacturer_TblRefID** (int): FOREIGN KEY to `m_device_manufacturer`. Stores the manufacturer of the smart meter.
- **NetworkLookup_TblRefID** (int): FOREIGN KEY to `l_network_lookup`. Maps the smart meter to its parent network (generally the DTR Network).
- **OrganisationLookup_TblRefID** (int): FOREIGN KEY to `l_organisation_lookup`. Maps the smart meter to its specific organizational hierarchy.
- **MF** (float): Stores the multiplying factor (MF) of the smart meter.
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates if the meter is currently in an active state.

- **IsNetMeter** (bit, Enum: `0`='No', `1`='Yes'): Indicates whether the smart meter functions as a net meter.
- **netmeter_change_ts** (datetime): Date and time when the net meter function was enabled in the smart meter.
- **InstallationDate** (datetime): Date and time when the smart meter was physically installed.
- **MeterType_TblRefID** (int, Enum: `1`='feeder meter', `2`='DT meter', `3`='consumer meter'): FOREIGN KEY to `m_meter_type`. Defines the physical placement or type of the smart meter.

Table: l_network_lookup

Description

This table contains the records for all electrical network entities (such as Substations, Feeders, and Distribution Transformer (DTR) networks). It defines the strict physical hierarchy of the grid, including how lower-level networks connect to higher-level networks and which utility organizations manage them.

Columns

- **NetworkLookup_TblRefID** (int): PRIMARY KEY. Unique internal identity number for each network entity.
- **Network_Code** (varchar): Standard identifier code for the network.
- **Network_RAPDRPCode** (varchar): RAPDRP-specific identifier code for the network.
- **Network_Name** (nvarchar): Name of the network entity.
- **Network_Address** (nvarchar): Physical address or location of the network entity.
- **HigherNetwork_ID** (int): FOREIGN KEY to `l_network_lookup` (Self-referencing). Maps the network to its parent or higher-level network entity.
- **NetworkHierarchy_TblRefID** (int, Enum: `1`='substation', `2`='feeder', `3`='dtr'): FOREIGN KEY to `m_network_hierarchy`. Defines the type of network and its level in the physical hierarchy.
- **OrganisationLookup_TblRefID** (int): FOREIGN KEY to `l_organisation_lookup`. Maps the network to the specific organization or office that manages it.
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates if the network entity is currently active.

- **SUPPLY_VTG_TblRefID** (int, Enum: `1`='11kv', `2`='33kv'): Stores the supply voltage capacity for the network.

Table: l_organisation_lookup

Description

This table contains the records for all utility organizations and offices (such as Circle, Division, Subdivision, and Section). It defines the strict hierarchy of these offices and how they nest within one another.

Columns

- **OrganisationLookup_TblRefID** (int): PRIMARY KEY. Unique internal identity number for each organization/office.
- **Office_Code** (varchar): Unique code assigned to the organization or office.
- **Office_Name** (nvarchar): Name of the organization or office.
- **HigherOffice_ID** (int): FOREIGN KEY to `l_organisation_lookup` (Self-referencing). Maps the organization to its parent or higher-level organization.
- **OrganisationHierarchy_TblRefID** (int): FOREIGN KEY to `m_organisation_hierarchy`. Stores the organization type and determines its level in the hierarchy (e.g., 1 being the top-level hierarchy, followed by subsequent numbers for lower levels).
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates if the organization is currently in an active state.

Table: m_connection_category

Description

This master lookup table defines the specific tariff codes and categorizations applied to consumers. It acts as the source of truth for classifying connections into business groups such as Domestic, Commercial, Agriculture, and Industrial.

Columns

- **ConnectionCategory_TblRefID** (int, Enum: `1`='KJ', `2`='KJ_BPL_MTR', `3`='DS1', `4`='DS1D', `5`='DS2D', `6`='DS3D', `7`='NDS1', `8`='NDS-IID(A)', `9`='NDS2D', `10`='LTIS1D', `11`='LTIS2D', `12`='IAS1', `13`='IAS2', `14`='PWWD', `15`='HGN', `16`='SS1D', `17`='LTEV', `18`='NDS1D', `19`='IAS2D'): PRIMARY KEY. Unique identifier for the connection category.

- **ConnectionCategory_Code** (varchar): UNIQUE. The standard shorthand code for the connection category (matches the ID enums).
- **ConnectionCategory_Desc** (varchar, Enum: `Kuteer Jyoti`, `Domestic`, `Commercial`, `LT Industrial`, `Agriculture`, `Public Water Works`, `Har Ghar Nal`, `Street Light`, `EV Charging`): The full descriptive name and classification of the connection category.

Table: m_connection_status

Description

This master lookup table defines the available connectivity statuses for a consumer's smart meter. It is used to filter consumers who are currently receiving power versus those who have been temporarily or permanently disconnected.

Columns

- **ConnectionStatus_TblRefID** (smallint, Enum: `1`='Connected', `2`='Disconnected', `3`='Permanent Disconnection'): PRIMARY KEY. Unique identifier for the connectivity state.
- **ConnectionStatus_Name** (varchar, Enum: `Connected`, `Disconnected`, `Permanent Disconnection`): The full descriptive name of the connectivity state.

Table: m_device_manufacturer

Description

This master lookup table stores the complete list of authorized manufacturers and brands for smart meters and related hardware devices in the system.

Columns

- **DeviceManufacturer_TblRefID** (int, Enum: `1`='UTILITY', `2`='ACS', `3`='ADME', `4`='AEW', `5`='AI', `8`='AMTL', `9`='ANSHU', `10`='ANZU', `11`='ARON', `12`='ASSLTO', `13`='AT', `14`='AVON', `15`='AYONPCFR', `16`='BAJAJ', `17`='BEM', `18`='B-E-M', `19`='BENTEC', `20`='BENTEK', `21`='BENTEX', `22`='BHEL', `23`='BHM', `24`='BM', `25`='BPL', `26`='C and H', `27`='CLASS', `28`='DASS', `29`='DCD', `30`='DH', `31`='DLMS', `32`='DTRMETER', `33`='DUKE', `34`='DUMMY', `35`='EAW', `36`='EC', `37`='ECE', `38`='ELYMER', `39`='EMCO', `40`='EMLT', `41`='FERRANTI', `42`='FL', `43`='FLAP', `44`='FLASH', `45`='FLESH', `46`='FNUS', `47`='GE', `48`='GENUS', `49`='GM', `50`='GPI', `51`='GTIS', `52`='HAITEK', `53`='HARIYANA', `54`='HARYANA', `55`='HAVELLS', `56`='HEV', `57`='HI', `58`='HM', `59`='HPL', `60`='HSM',

`61`='HT', `62`='HYTECH', `63`='IB', `64`='IGM', `65`='IM', `66`='INDIA', `67`='INDO FRANC', `68`='INDOTECH', `69`='IT', `70`='JAIPUR', `71`='JS,Baroda', `72`='L and G', `73`='LANDIS', `77`='LASEN', `78`='LENKAR', `81`='LINKWELL', `83`='LPT', `85`='LW', `86`='M and I', `87`='MADRAS', `88`='MANGAL', `89`='MBM', `90`='MGI', `91`='MI', `92`='MIG', `93`='MONTAIL', `94`='MONTEL', `95`='MONTEX', `96`='MOTRAL', `97`='MPVY', `98`='NACODA', `99`='NAK', `100`='NAKODA', `101`='NAMTECH', `102`='NEL', `103`='NKD', `104`='NM', `105`='PAL', `106`='PALMOAN', `107`='PALMOHA', `108`='PALMOHAN', `109`='PM', `110`='QTY', `111`='RC', `112`='REKKON', `113`='REMCO', `114`='RIKKEN', `115`='S', `116`='SCHLUMBERG', `120`='SECURE', `121`='SEMI COND', `123`='SIMCO', `124`='SINTEX', `125`='SOCOMEC', `126`='SOMAC', `128`='TGL', `129`='TLG', `130`='TOBRA', `131`='TRINCRGN', `132`='TTL', `133`='TURBO', `134`='UE', `135`='UNILAC', `136`='UNILEC', `137`='UNILIC', `138`='UNITED', `139`='UNIVERSAL', `140`='UNMETERED', `141`='UTAKE', `142`='VINOTEK', `143`='VISION', `144`='VISIONTAK', `145`='VISIONTAKE', `146`='VISIONTEC', `147`='VISIONTEK', `148`='VISIONTEX', `149`='VISNTEC', `150`='VISONTEK', `151`='VISTEC', `152`='VISTON', `153`='VS', `154`='VT', `155`='VTEK', `156`='VXL', `157`='VZS', `158`='WAATHOUR', `159`='WINNER', `160`='WPANLRAF', `162`='SARAL', `163`='GENRETION', `166`='L&T', `167`='Ametek', `168`='EL SEWEDY', `169`='GENTEK', `170`='INDIA PVT LTD', `171`='MAXWELL INDIA', `172`='PAL MOHAN', `173`='PLUTUS', `174`='POWER TECH', `175`='RMC INDIA', `176`='SEMI CONDUCTOR', `177`='SUNSTAR', `178`='VISION TECH', `179`='SCHNEIDER', `180`='OMNI AGATE', `181`='ISKRAEMECO', `261`='SWITCH GEAR', `262`='SEIPL', `263`='Capital', `301`='Kimbal', `302`='Sona Electricals Bhopal', `303`='Vishal Transformer and Switch gears'): PRIMARY KEY. Unique identifier for the device manufacturer.
- **Manufacturer_Name** (nvarchar): The full descriptive name of the device manufacturer.
- **isActive** (bit, Enum: `0`='No', `1`='Yes'): Indicates if the manufacturer is currently active and approved for use.

Table: m_hes

Description

This master lookup table defines the various Head-End Systems (HES) integrated into the MDMS. The HES is the central software that communicates directly with the smart meters to collect readings and push configurations.

Columns

- **HESID** (smallint, Enum: `1`='Scaler', `2`='LT AMR', `3`='HT AMR', `4`='HES-Cyan', `5`='Esya', `6`='HES-BCITs', `7`='HES-Genus', `8`='Crystal-2', `9`='LnG', `10`='Schneider-HES', `11`='Genus-HES', `12`='Genus-HES2', `13`='Ashoka-HES', `14`='BCITS- HES2', `15`='Ashoka-HES2'): PRIMARY KEY. Unique identifier for the Head-End System.
- **HESNAME** (varchar): The descriptive name of the Head-End System software or vendor.
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates whether this Head-End System is currently active and supported in the environment.

Table: m_network_hierarchy

Description

This master lookup table defines the specific tiers within the electrical grid's physical hierarchy. It is used to categorize network entities into top-level Substations, mid-level Feeders, and edge-level Distribution Transformers (DTRs).

Columns

- **NetworkHierarchy_TblRefID** (int, Enum: `1`='Sub Station', `2`='Feeder', `3`='DTR'): PRIMARY KEY. Unique identifier for the network hierarchy level.
- **NetworkHierarchy_Code** (varchar): Standard numeric code representing the hierarchy level.
- **NetworkHierarchy_Name** (nvarchar, Enum: `Sub Station`, `Feeder`, `DTR`): The descriptive name of the network tier.
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates whether this network hierarchy level is currently active in the system.

Table: m_organisation_hierarchy

Description

This master lookup table defines the specific tiers within the utility's organizational structure. It is used to categorize offices into a strict hierarchy, where `1` represents the highest level (Headquarters) and subsequent numbers represent lower operational branches down to the Section level.

Columns

- **OrganisationHierarchy_TblRefID** (int, Enum: `1`='HQ', `2`='Circle', `3`='Division', `4`='Subdivision', `5`='Section'): PRIMARY KEY. Unique identifier for the organizational hierarchy level.

- **OrganisationHierarchy_Code** (varchar): Standard code representing the organizational level.
- **OrganisationHierarchy_Name** (varchar, Enum: `HQ`, `Circle`, `Division`, `Subdivision`, `Section`): The descriptive name of the organizational tier.
- **IsActiveStatus** (bit, Enum: `0`='No', `1`='Yes'): Indicates whether this organizational hierarchy level is currently active in the system.

Table: m_paymenttype_contract

Description

This master lookup table defines the available payment contract types for consumers. It is primarily used to distinguish between Prepaid and Postpaid accounts across the system.

Columns

- **PaymentContract_TblRefID** (int, Enum: `1`='Postpaid', `2`='Prepaid'): PRIMARY KEY. Unique identifier for the payment contract type.
- **PaymentContract_Name** (varchar, Enum: `Postpaid`, `Prepaid`): The descriptive name of the payment contract.

Table: m_servicepoint_meterphase

Description

This master lookup table defines the available physical phase configurations and connection types for smart meters across the system. It helps differentiate between standard single-phase residential meters, three-phase commercial/industrial meters, and High Tension (HT) connections.

Columns

- **ServicePointMeterPhase_TblRefID** (int, Enum: `1`='1 PH', `2`='3PH WC', `3`='3PH LT CT', `4`='HT', `6`='3PH LT CT'): PRIMARY KEY. Unique identifier for the meter phase configuration.
- **MeterPhase_Name** (varchar, Enum: `1 PH`, `3PH WC`, `3PH LT CT`, `HT`): The standard descriptive name of the meter phase.
- **IsActive** (bit, Enum: `0`='No', `1`='Yes'): Indicates whether this phase configuration is currently active in the system.

Table: m_sl_type

Description

This master lookup table defines the units of electrical power measurement used to represent a consumer's Sanctioned Load or Contract Demand. It ensures standardized calculations across different connection types.

Columns

- **SL_TYPE_ID** (smallint, Enum: `1`='KW', `2`='KVA', `3`='HP', `4`='W'): PRIMARY KEY. Unique identifier for the sanctioned load unit of measurement.
- **SL_TYPE_NAME** (varchar, Enum: `KW`, `KVA`, `HP`, `W`): The standard abbreviation/name of the electrical power unit.

Relationship Map – mdms_master

1. Consumer Core Joins

The central entity is `l_consumer_lookup` (Alias: lcl).

- **Target:** `l_meter_lookup` (Alias: lml)
  - **Join Logic:** `lcl.MeterLookup_TblRefId = lml.MeterLookup_TblRefID`
  - **Purpose:** Links one consumer to their primary installed smart meter.

- **Target:** `m_connection_status` (Alias: mcs)
  - **Join Logic:** `lcl.ConnectionStatus_TblRefID = mcs.ConnectionStatus_TblRefID`
  - **Purpose:** Gets the physical connectivity/relay status.

- **Target:** `m_connection_category` (Alias: mcc)
  - **Join Logic:** `lcl.ConnectionCategory_TblRefID = mcc.ConnectionCategory_TblRefID`
  - **Purpose:** Gets the consumer's tariff/category (e.g., Domestic, Commercial).

- **Target:** `m_paymenttype_contract` (Alias: mptc)
  - **Join Logic:** `lcl.PaymentContract_TblRefID = mptc.PaymentContract_TblRefID`
  - **Purpose:** Identifies if the consumer is Prepaid or Postpaid.

- **Target:** `m_sl_type` (Alias: mslt)
  - **Join Logic:** `lcl.SL_TYPE_ID = mslt.SL_TYPE_ID`
  - **Purpose:** Gets the unit of measurement for sanctioned load.

2. Meter Hardware & Mapping Joins

These joins originate from `l_meter_lookup` (Alias: lml) to define hardware specs and physical locations.

- **Target:** `m_hes` (Alias: mhes)
  - **Join Logic:** `lml.HESID = mhes.HESID`
  - **Purpose:** Maps the meter to its communicating Head-End System.
- **Target:** `m_servicepoint_meterphase` (Alias: msmp)
  - **Join Logic:** `lml.ServicePointMeterPhase_TblRefID = msmp.ServicePointMeterPhase_TblRefID`
  - **Purpose:** Gets the physical phase (1 Phase, 3 Phase, etc.).
- **Target:** `m_device_manufacturer` (Alias: mdm)
  - **Join Logic:** `lml.DeviceManufacturer_TblRefID = mdm.DeviceManufacturer_TblRefID`
  - **Purpose:** Identifies the smart meter's brand/manufacturer.
- **Target:** `l_network_lookup` (Alias: lnl)
  - **Join Logic:** `lml.NetworkLookup_TblRefID = lnl.NetworkLookup_TblRefID`
  - **Purpose:** Maps the smart meter to its parent network (usually a DTR).
- **Target:** `l_organisation_lookup` (Alias: l_org)
  - **Join Logic:** `lml.OrganisationLookup_TblRefID = l_org.OrganisationLookup_TblRefID`
  - **Purpose:** Maps the smart meter to its managing utility office.

3. Hierarchy & Self-Joins (Network & Organisation)

These tables map the grid and utility structure, using self-joins for parent-child navigation.

- **Network to Organisation:**
  - **Join Logic:** `l_network_lookup.OrganisationLookup_TblRefID = l_organisation_lookup.OrganisationLookup_TblRefID`

- **Network Master Lookups:**
  - **Join Logic:** `l_network_lookup.NetworkHierarchy_TblRefID = m_network_hierarchy.NetworkHierarchy_TblRefID`
- **Network Self-Join (Parent/Child):**
  - **Join Logic:** `child_net.HigherNetwork_ID = parent_net.NetworkLookup_TblRefID`
  - **Purpose:** E.g., Finding the Feeder for a specific DTR.
- **Organisation Master Lookups:**
  - **Join Logic:** `l_organisation_lookup.OrganisationHierarchy_TblRefID = m_organisation_hierarchy.OrganisationHierarchy_TblRefID`
- **Organisation Self-Join (Parent/Child):**
  - **Join Logic:** `child_org.HigherOffice_ID = parent_org.OrganisationLookup_TblRefID`
  - **Purpose:** E.g., Finding the Division for a specific Subdivision.

4. Important Implementation Rules

1. **Self-Referencing Aliases:** When writing SQL that traverses hierarchies, you MUST use distinct aliases for the parent and child tables (e.g., `l_organisation_lookup child_org` JOIN `l_organisation_lookup parent_org`).
2. **Enum Optimization:** Because master lookups like `m_connection_status`, `m_paymenttype_contract`, and `m_hes` have their ID values documented directly as `Enum` lists in the core table OKFs, **do not execute JOINs to these master tables just for basic `WHERE` filtering.** Only JOIN them if you need to `SELECT` the string name in the output.

5. Hierarchy Traversal Guide

When writing queries that require navigating up or down the organizational or network trees, strictly follow these structural chains. The `OrganisationHierarchy_TblRefID` and `NetworkHierarchy_TblRefID` determine the exact level.

A. Organisation Chain (Top to Bottom)

1. **HQ** (`OrganisationHierarchy_TblRefID = 1`) -> *Highest Parent*
2. **Circle** (`OrganisationHierarchy_TblRefID = 2`)

3. **Division** (`OrganisationHierarchy_TblRefID = 3`)
4. **Subdivision** (`OrganisationHierarchy_TblRefID = 4`)
5. **Section** (`OrganisationHierarchy_TblRefID = 5`) -> *Lowest Child*

B. Network Chain (Top to Bottom)

1. **Sub Station** (`NetworkHierarchy_TblRefID = 1`) -> *Highest Parent*
2. **Feeder** (`NetworkHierarchy_TblRefID = 2`)
3. **DTR** (`NetworkHierarchy_TblRefID = 3`) -> *Lowest Child*
   _Note: Smart Meters are generally mapped to the DTR level via `l_meter_lookup.NetworkLookup_TblRefID`.

Example of traversing up from Section to Division:

```
SELECT section.Office_Name, division.Office_Name
FROM l_organisation_lookup section
-- Join up to Subdivision
JOIN l_organisation_lookup subdiv ON section.HigherOffice_ID =
subdiv.OrganisationLookup_TblRefID
-- Join up to Division
JOIN l_organisation_lookup division ON subdiv.HigherOffice_ID =
division.OrganisationLookup_TblRefID
WHERE section.OrganisationHierarchy_TblRefID = 5;
```